

AI Foundation Models in Biomedicine

Tutorial 2

Winter Semester 2025/26

Task 1: Installation and Environment Setup

To set up the cell2sentence environment, follow the steps below:

1. Clone the Repository

Pull the repository locally using:

```
1 git clone https://github.com/vandijklab/cell2sentence.git
```

Navigate a terminal into the root of the repository.

2. Create an Anaconda Environment

Create an Anaconda environment using Python 3.8 with:

```
1 conda create -n cell2sentence python=3.8
```

3. Activate the Environment

Activate the new environment:

```
1 conda activate cell2sentence
```

4. Install the Package

Run the setup using:

```
1 make install
```

Task 2: Run the Tutorial Notebooks

Navigate to the tutorial section of the repository and run the following notebooks in order.

Hint: Some tasks in the tutorials (e.g., inference on full datasets) may be computationally expensive. To speed up experimentation, you can limit the number of processed cells by selecting a subset of the dataset like this:

```
1 test_ds = arrow_ds.select(range(100))
2 test_ds
```

or

```
1 adata_small = adata[:100].copy() # subset the AnnData
2 adata_small.obsm["c2s_cell_embeddings"] = embedded_cells
```

You may skip all model finetuning steps in the tutorials. Instead, use the pretrained models provided in the *C2S Model Zoo*. These models are available on HuggingFace and can be loaded directly using:

```
1 cell_type_prediction_model_path = "vandijklab/C2S-Pythia-410m-
  cell-type-prediction"
2 save_dir = "/path/to/save/dir"
3 save_name = "cell_type_pred_pythia_410M_inference"
4 csmodel = cs.CSModel(
5     model_name_or_path = cell_type_prediction_model_path,
6     save_dir = save_dir,
7     save_name = save_name,
8 )
```

Since you are not finetuning models, the dataset split loading code in the tutorials can be ignored or commented out:

```
1 # base_path = "/".join(cell_type_prediction_model_path.split("/")
  #[:-1])
2 # with open(os.path.join(base_path, 'data_split_indices_dict.pkl
  #'), 'rb') as f:
3 #     data_split_indices_dict = pickle.load(f)
4 # data_split_indices_dict.keys()
5 # len(data_split_indices_dict["train"])
6 # len(data_split_indices_dict["val"])
7 # len(data_split_indices_dict["test"])
```

Notebooks to run:

- c2s_tutorial_0_data_preparation.ipynb
- c2s_tutorial_1_rank_transformation_and_reconstruction.ipynb
- c2s_tutorial_2_cell_embedding.ipynb
- c2s_tutorial_4_cell_type_prediction.ipynb

Further Documentation

For more detailed information on installation, usage, and advanced features, please refer to the official `cell2sentence` documentation:

<https://vandijklab-cell2sentence.readthedocs.io/en/latest/>